

NDNSF-UAV: Map-Selected Multi-UAV Inspection

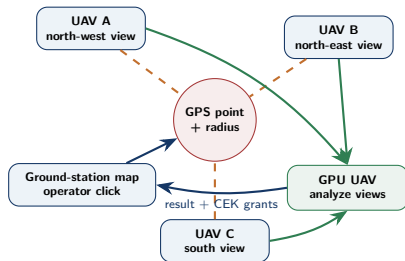
Late-bound camera UAVs capture complementary views; a selected GPU UAV analyzes them and returns named results

WORK IN PROGRESS

DATA-DRIVEN

SIMULATION TO FIELD

Concrete Case: Capture, Analyze, and Display One GPS Point



One concrete task

- ▶ The operator clicks a location; the planner selects three camera roles and one GPU-capable Analyzer.
- ▶ Each camera publishes one encrypted `ViewImage`; the Analyzer fetches admitted objects by name and processes them.
- ▶ The Analyzer returns annotations and content-encryption-key (CEK) grants; the Requester fetches the original images and overlays the results.

Proposed first case, not a field demonstration

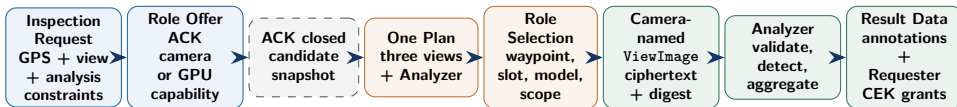
A known location narrows the search; pointing, visibility, pose error and cross-view correspondence still need validation.

Key point

Reuse each immutable image at both Analyzer and Requester; return annotations and recipient-bound key grants.

Evidence: Multi-UAV visual inspection and coverage systems already distribute image-acquisition work across aircraft [1,2].

Proposed NDNSF Workflow: Physical Execution Still Needs Feedback



Request and planning

- Request: point/radius, time window, view diversity, image quality, model requirement, and result deadline.
- Offer: cameras report reachability and view capability; processors report GPU, model, queue state, and input limits.
- Plan: View-A/B/C plus one Analyzer; named edges connect each View role to the Analyzer.

Execution and result admission

- Selection assigns viewpoints and binds the Analyzer, Requester, named dependencies, model, and deadline.
- Each Camera publishes once under `/<camera>/NDNSF/COLLAB/...`, with request/plan/attempt, capture metadata, and a CEK wrapped to the Analyzer.
- The Analyzer requires prefix, signature, Plan role-Provider, and dependency-edge agreement. Its own signed result names every image/digest and carries annotations, failures, and the same CEKs rewrapped to the authorized Requester.

ACK SNAPSHOT

ONE PLAN

REUSED VIEW DATA

RESULT + KEY GRANT

EXPLICIT FAILURES

The Requester fetches original images from cache or Repo, decrypts them, and overlays the returned annotations.

Evidence: Current API: ACK closure and plan commit. In progress: cross-role input/result enforcement.

Practical Boundary, MVP, and Required Evidence

Responsibility boundary

- ▶ **NDNSF**: protected Request, discovery, ACK closure, producer-owned names, signer/role/dependency admission, recipient-bound CEK grants, and failure state
- ▶ **UAV application**: viewpoint/navigation, capture metadata, analysis invocation, annotation schema, and Requester-side overlay
- ▶ **Perception backend**: existing per-view detector and deterministic evidence aggregation; it does not decide Provider assignment
- ▶ **Autopilot and onboard safety**: stabilization, readiness, geofence and final rejection; collision avoidance requires configured sensing and planning

Current implementation work. Complete dependency/result admission and Requester-bound CEK grants; integrate frame metadata and annotations.

Target claim to validate: publish each image object once, analyze it remotely, and securely reuse it at the Requester.

First measurable MVP

- ▶ one static outdoor GPS point, three camera roles, three camera UAVs, one GPU Analyzer, and one existing detector
- ▶ test provider unavailability, mobility, Data loss, missed deadlines, and unsafe-assignment rejection

Measurements and ablations

- ▶ completion, valid-view and analysis success, latency, bytes, image publications per capture, and flight/energy proxy
- ▶ one versus three views under occlusion; nearest versus geometry-aware selection; Analyzer failure and retry

Claim boundary: no new recognition algorithm, moving-target identity, fleet autonomy, triangulation, or 3-D accuracy claim.

Evidence: [1] Mansouri et al., Control Engineering Practice 2018; [2] Luna et al., Journal of Field Robotics 2024; [3] MAVLink Common Message Set.

Simulation Can Exercise the Workflow; It Cannot Prove Capture

Boundary	Simplified case	Real deployment needs
Vehicle position	Supplied position and reachability	GPS/GNSS + IMU pose, timestamps, uncertainty and stale-state rejection
Camera viewpoint	Assigned angle or supplied image	Gimbal feedback, camera API, calibrated optics, exposure and capture confirmation
Target observation	Target assumed visible	Terrain/obstacle checks, image quality, target localization and cross-view association
Execution environment	Controlled timing and links	Flight dynamics, radio loss, battery/compute limits and recovery

Key point

Existing service or simulation tests do not establish that three aircraft captured the same physical target.

Evidence: Scope: current UAV README describes a CPU collaboration slice; real MiniNDN/SITL evidence is separate. This deck reports no physical-flight qualification.

Close the Physical Loop Before Publishing a Successful View

1. **Locate and plan:** target coordinates/height + aircraft pose → reachable viewpoint and camera line of sight.
2. **Move and point:** autopilot waypoint + gimbal command → observed arrival and settled camera orientation.
3. **Capture and check:** trigger camera → returned frame, exposure time, sharpness and target visibility.
4. **Publish or retry:** bind accepted image to measured pose, calibration, request/plan/attempt; otherwise retry within budget or report failure.

Key point

A command ACK confirms command handling, not a useful image. The service completes only after checking physical feedback and captured content.

Evidence: Candidate adapter APIs: MAVSDK Telemetry, Camera and Gimbal [4]. Firmware/device support must be verified. NDNSF handles service collaboration; flight stabilization stays onboard.

Localization and Multi-View Recognition Are Separate Tasks

- ▶ **Locate the aircraft:** fuse navigation sensors; record pose uncertainty. Define global/local coordinates and altitude reference.
- ▶ **Locate the observation:** calibrate camera intrinsics and camera–gimbal–body transforms; align image exposure with pose. A pixel ray needs depth, terrain, or multiple views to infer a world point [5].
- ▶ **Recognize across views:** detect per image, associate the same target, reject mismatches, then fuse evidence with uncertainty. Three bounding-box lists alone are not multi-view recognition.

Key point

First validate a surveyed static target with per-view results. Add cross-view fusion next; 3-D localization and moving-target tracking require additional algorithms and evidence.

Evidence: [5] OpenCV calibration/geometry primitives support geometric estimation; they do not provide a complete UAV perception pipeline.

Proposed integration, not an implemented accuracy claim.

Reach the Target Through Four Observable Stages

Stage	Evidence required before expanding scope
1. Sensor bench	Real camera/gimbal APIs, calibration and exposure/pose pairing; verify rejected commands, timeouts and missing frames.
2. Flight simulation	Software-in-the-loop (SITL), then hardware-in-the-loop (HITL) where supported [6]: close navigation/capture feedback; inject GPS error, occlusion and link loss.
3. One physical UAV	Surveyed static target; repeatable arrival, pointing and useful captures; compare measured location/latency against ground truth.
4. Multiple UAVs	Complementary real views and selected Analyzer; verify target association, fusion, bounded retries and loss of one participant.

Key point

Keep the same service contracts across stages. Promote a stage using recorded measurements, not successful message exchange alone.

Evidence: [6] PX4 distinguishes simulated flight stacks and hardware-backed simulation. Neither establishes real camera, radio or flight performance.
All four stages are proposed work here.

Define Success Before the Field Experiment

- ▶ **Freeze the operating envelope:** target size/range, altitude, lighting, wind and allowable location error; select hardware against it.
- ▶ **Freeze measurable thresholds:** pointing/position error, exposure–pose skew, valid-view rate, recognition precision/recall and end-to-end p95 latency. Values need bench data, not invented guarantees.
- ▶ **Compare on the same trials:** one versus multiple views, per-view results versus fusion; measure added traffic, capture delay and energy alongside accuracy.
- ▶ **Exercise failures:** stale GPS, gimbal saturation, missing frames, wrong-view association, Analyzer loss and deadline expiry. Expect bounded retries or explicit failure.

Key point

Immediate next deliverable: one real camera/gimbal capture with synchronized pose and independently checked target visibility. Then scale the collaboration.

References and Evidence Boundary

Hardware integration candidates; not claims of current integration

- 4 MAVSDK C++ guides: Telemetry, Camera, Gimbal. Gimbal guide specifies MAVLink Gimbal Protocol v2.
- 5 OpenCV 4.13: Camera Calibration and 3D Reconstruction.
- 6 PX4 Simulation and Hardware Simulation. Check the chosen board, simulator and firmware combination.

Existing context references retained from the example

- 1 Mansouri et al., *Control Engineering Practice*, 2018; [2] Luna et al., *Journal of Field Robotics*, 2024. Context only; not evidence of this implementation.
- 3 MAVLink Common Message Set. Protocol definitions do not certify device support.

Evidence: Official integration references checked 2026-09-10. Local scope: NDNSF-UAV-APP/README.md and the editable UPDATES.tex. No new hardware run or product qualification is reported.